

# Pharmacogenomic (PGx) Risk Dashboard

## 1 Frontend/Hosting



AWS S3

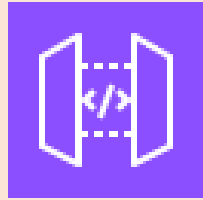


Route 53



CloudFront

## 2 API Gateway



## 3 AWS Lambda



## 4 Backend



AWS S3

- **Per-Density-Bin Models (up to 84)**  
**CatBoost + XGBoost + XGBoost-RF**  
**3 algorithms × 4 bins × 7 age bands (~469 MB)**
- **CPIC DB Snapshot (March 2026) 573 gene-drug pairs**
  - **Level A/B only 12 PGx genes (~48 MB)**
- **Feature Schemas + Training Medians Per-bin thresholds**
  - **Imputation of Normality (~18 MB)**